

Data Sources Used to Predict Changes in American Pika Habitat

To model changes in suitable habitat for the American pika (*Ochotona princeps*), multiple datasets describing species occupancy, habitat characteristics, topography, and climate were integrated. Preference was given to datasets that distinguish between **currently occupied and extirpated sites**, allowing models to capture habitat persistence and local extinction rather than relying solely on presence records.

1. Occupancy, Presence–Absence and Extirpation Datasets

- **Sierra Nevada / Southern Cascades Occupancy–Extirpation Dataset (2014–2019):** This U.S. Geological Survey (USGS) dataset provides one of the most comprehensive records of pika occupancy status in California. Researchers surveyed 150 talus patches, classifying each site into one of three categories: Current occurrence, Former occurrence (local extirpation), No evidence of pika occupancy. The dataset also includes geographic coordinates and elevation. Because it explicitly identifies occupied, extirpated and unoccupied habitat, it is particularly valuable for modelling the environmental conditions associated with both persistence and local extinction.

Millar, C. I., Westfall, R. D., Delany, D. L., et al. (2018). Ordinal occurrence states of American pikas in the northern Sierra Nevada and southern Cascade Range. U.S. Geological Survey.

<https://www.usgs.gov/data/ordinal-occurrence-states-american-pikas-northern-sierra-nevada-and-southern-cascade-range>

- **Northern Rockies / Cascades Persistence Survey (2015–2017):** USGS scientists revisited historical pika locations documented before 1980 across Montana, Wyoming, Idaho, Oregon, and Washington to determine whether populations had persisted or disappeared. At each survey location, researchers collected: Occupancy status and Habitat characteristics.

U.S. Geological Survey ScienceBase. (2025). American pika persistence surveys in the northern Rocky Mountains and Cascade Range.

<https://www.sciencebase.gov/catalog/item/682e9ccad4be020ae7e1a205>

- **Great Basin Presence–Absence and Status Dataset:** To improve geographic coverage, presence and absence information was also obtained from the study **Distribution, climatic relationships, and status of American pikas (*Ochotona princeps*) in the Great Basin, USA** (Millar et al., 2018). This study compiled one of the largest occurrence datasets for American pikas in the Great Basin by integrating historical records with extensive field surveys conducted since 2005. The dataset spans approximately **40 mountain ranges** and an elevational range of **1,631–4,009 m**, representing a much broader climatic range than previously recognised. This dataset is particularly valuable because it provides both **presence and absence/extirpation information**, enabling habitat models to distinguish between

environmental conditions associated with population persistence, local extinction and unsuitable habitat rather than relying solely on presence-only observations.

Millar, C. I., Delany, D. L., Hersey, K. A., Jeffress, M. R., Smith, A. T., Van Gunst, K. J., & Westfall, R. D. (2018). *Distribution, climatic relationships, and status of American pikas (Ochotona princeps) in the Great Basin, USA. Arctic, Antarctic, and Alpine Research*, 50(1), e1436296.

<https://www.tandfonline.com/doi/full/10.1080/15230430.2018.1436296#d1e2787>

- **Global Biodiversity Information Facility (GBIF) Occurrence Records. American pika occurrence records:** GBIF is an international, open-access biodiversity database that aggregates species occurrence records from museums, research institutions, government agencies, and citizen science programs worldwide. For this study, American pika presence records were used to supplement field survey datasets, expanding the geographic coverage of known occurrences across the species' range. GBIF provides georeferenced records of confirmed observations but generally does not include true absence or extirpation data, making it most useful for complementing occupancy datasets and validating species distributions.

https://www.gbif.org/occurrence/search?occurrence_status=present

2. Topographic Variables

Topography strongly influences pika habitat through its effects on temperature, moisture and snow persistence. Two terrain variables were included:

- **Elevation (m):** obtained directly from source datasets where available or derived from geographic coordinates using the Open-Elevation API.
- **Slope (degrees):** calculated from elevation differences between neighbouring locations (~90 m spacing), providing a consistent estimate of terrain steepness.

Open-Elevation API. <https://www.open-elevation.com>

3. Climate Variables

Climate predictors were obtained primarily from **WorldClim Version 2.1**, which provides high-resolution (~1 km) 30-year climate normals describing long-term average climate conditions derived from weather station observations. Six bioclimatic variables with well-established ecological relevance to pikas were selected:

Variable	Description
BIO1	Mean annual temperature (Celsius)
BIO4	Temperature seasonality : Standard Deviation (Monthly Mean Temperature)×100 - reflects how much temperatures fluctuate between seasons. High seasonality is often associated with cold, snowy winters and relatively warm summers , conditions typical of many alpine environments.

- BIO5 Maximum temperature of the warmest month (Celsius)
- BIO6 Minimum temperature of the coldest month (Celsius)
- BIO12 Annual precipitation (mm)
- BIO15 **Precipitation seasonality: Standard Deviation of Monthly Precipitation/ Mean Monthly Precipitation ×100** - it captures precipitation variability, rather than the total amount of precipitation (BIO12). Precipitation seasonality can influence habitat suitability in several ways:
- **Snowpack formation:** Pikas rely on persistent winter snow to insulate their talus habitats from extreme cold. Changes in when and how precipitation falls (snow versus rain) can affect winter survival.
 - **Vegetation productivity:** Seasonal precipitation influences the growth of grasses, forbs, and shrubs that pikas collect to build their haypiles. Highly seasonal precipitation may shorten or shift the growing season.
 - **Drought stress:** Regions with prolonged dry periods may experience reduced plant biomass, leaving pikas with fewer food resources.
 - **Climate refugia:** Areas with more stable precipitation regimes may better support alpine vegetation and provide more reliable habitat under climate change.

Maximum summer temperature (BIO5) is particularly important because American pikas experience severe heat stress and may not survive prolonged exposure to temperatures above approximately **25.5°C** without access to cool talus refuges.

Additional climate variables were obtained from **TerraClimate** to capture historical climate variability from **1950 onwards**, including **snow water equivalent, drought index, and annual maximum and minimum temperatures**. These variables complement the long-term climate normals provided by WorldClim by incorporating interannual variability in temperature, drought, and snowpack, all of which are recognised as important drivers of American pika persistence and habitat suitability. TerraClimate is a high-resolution (~4 km) global dataset that provides monthly climate variables for terrestrial surfaces, making it well suited for ecological applications that require long-term, spatially explicit climate data.

Abatzoglou, J. T., Dobrowski, S. Z., Parks, S. A., & Hegewisch, K. C. (2018). TerraClimate, a high-resolution global dataset of monthly climate and climatic water balance from 1958–2015. Scientific Data, 5, 170191.

Fick, S. E., & Hijmans, R. J. (2017). WorldClim 2: New 1-km spatial resolution climate surfaces for global land areas. International Journal of Climatology, 37, 4302–4315. <https://www.worldclim.org/data/index.html>

<https://www.climatologylab.org/terraclimate.html>

4. Limitations

An possible limitation of our environmental dataset is that reliable, high-resolution information describing talus characteristics (e.g., rock size, depth, and structural complexity) and local vegetation composition was not available for all survey locations. Consequently, these habitat attributes could not be included as predictor variables in the modelling process. Nevertheless, this limitation is expected to have a relatively limited impact on the study because the occurrence records were obtained from locations that have been the subject of targeted American pika field surveys. These sites are generally situated within alpine or subalpine environments that already contain the rocky talus habitats and associated vegetation communities known to support pika populations. Therefore, although local differences in talus structure and plant species composition undoubtedly exist, it is reasonable to assume that the surveyed locations share broadly similar habitat characteristics to established pika refugia. The environmental predictors included in this study consequently focus on the climatic and topographic factors that are expected to drive variation in habitat suitability among these otherwise comparable mountain environments.

5. Future Climate Projections (@OLGA if you can add/correct/check if the projection from the 2 different datasets are compatible? Depending on the model chosen for the bio -variables)

To assess the potential impacts of climate change on the distribution of the American pika, the trained model was applied to projected future environmental conditions. Rather than predicting habitat suitability only under present-day climate, the same set of environmental predictors used during model training was generated using future climate projections for each survey location. This approach ensures consistency between the training and prediction datasets while allowing the models to estimate how habitat suitability may change under different levels of future warming.

By comparing current and projected habitat suitability, it is possible to identify areas where environmental conditions are expected to remain suitable, areas where habitat quality is likely to deteriorate, and regions where pika populations may face an increased risk of local extinction. These projections also provide insights into potential range shifts and the persistence of climate refugia under future climate scenarios.

Because the environmental predictors originate from two different climate datasets, future projections were obtained separately for each data source and then combined to create the complete set of predictors required by the model.

Future Projections for TerraClimate-Derived Variables: Future projections for the TerraClimate-derived variables were obtained from climate datasets developed using a pattern-scaling approach. Two future warming scenarios were considered, representing global mean temperature increases of 2°C and 4°C relative to pre-industrial conditions. These scenarios provide a consistent framework for evaluating how increasing levels of global warming may affect pika habitat suitability.

The future climate datasets were generated using monthly projections from 20 CMIP6 Global Climate Models (GCMs) following the methodology described by Qin et al. (2020). Rather than relying on a single emissions pathway or individual climate model, the pattern-scaling approach estimates future monthly climate by applying projected changes in both climatic means and variability as a function of global temperature increase. This methodology is computationally efficient while producing projections that are largely independent of any particular greenhouse gas emissions scenario or individual climate model, making it well suited for ecological impact assessments based on global warming thresholds.

The projected TerraClimate variables include future temperature, precipitation, drought, and snow-related metrics, allowing the model to evaluate how both average climatic conditions and climatic extremes may influence future habitat suitability.

Future Projections for WorldClim-Derived Variables: Future projections for the WorldClim-derived bioclimatic variables were obtained from WorldClim v2.1 future climate datasets, which provide downscaled climate projections generated from the latest generation of CMIP6 Global Climate Models. These datasets use the same spatial resolution and bioclimatic variable definitions as the historical WorldClim dataset employed during model training, ensuring direct compatibility between current and future predictors.

The future bioclimatic variables describe projected long-term climatic conditions, including annual temperature, precipitation, seasonality, and climatic extremes, under different greenhouse gas emission pathways. By replacing the historical WorldClim variables with their projected counterparts while keeping the remaining environmental predictors consistent, the habitat suitability models can estimate how the climatic niche of the American pika is expected to shift under future climate conditions.

Combining future projections from both TerraClimate and WorldClim enables the models to capture changes in long-term climatic patterns as well as interannual climate characteristics that are known to influence pika ecology. The resulting habitat suitability maps provide a spatially explicit assessment of how climate change may alter the distribution of suitable habitat, identify potential climate refugia, and highlight regions where conservation efforts may be most critical under future warming scenarios.

*Qin, Y., Abatzoglou, J. T., Siebert, S., Huning, L. S., AghaKouchak, A., Mankin, J. S., Hong, C., Tong, D., Davis, S. J., & Mueller, N. D. (2020). Agricultural risks from changing snowmelt. *Nature Climate Change*.*